Intelligent Forecasting: An AI-Powered Tool for Predicting Startup Revenue in Venture Capital

Technical Report (10–15 pages)

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0.1 Executive Summary

Venture capital (VC) investment decisions are made under extreme uncertainty. Early-stage companies often lack stable financial histories, and the available data is sparse, heterogeneous, and biased toward specific geographies and sectors. This project delivers a **production-oriented forecasting system** that predicts a startup's **next four quarters of ARR (Annual Recurring Revenue)** using a LightGBM-based model, while explicitly addressing the core constraint of real-world usage: **users cannot provide 100+ structured features** at prediction time.

The shipped system consists of:

- **Tier-based input contract**: users provide a minimal set of inputs (Tier 1) and optional advanced metrics (Tier 2).
- **Intelligent Feature Completion**: the system constructs a realistic full feature vector by identifying similar companies in the historical dataset and imputing missing metrics using robust statistics.
- **Hybrid forecasting**: a routing layer detects non-standard trajectories (decline, stagnation, volatility) and falls back to a transparent **Rule-Based Health Assessment** that produces explainable projections using industry benchmarks.
- **FastAPI service**: the system is exposed via production-ready API endpoints for single-company forecasts, batch CSV processing, chat-based explanation, and macro context.

On a held-out test set, the LightGBM model achieves **overall (R^2)**. Performance varies by horizon (e.g., (R^2) for the nearest quarter and (R^2) for the fourth quarter ahead), which is expected for multi-step forecasting in volatile startup environments. Predictions are returned with a pragmatic **±10% decision-support range** to help VC users reason about risk and scenario bounds.

0.2 1. Problem Statement and Success Criteria

0.2.1 1.1 Prediction task

Given a company's recent ARR trajectory (four quarters), headcount, and sector (plus optional Tier 2 metrics), forecast:

- **Q+1 to Q+4 ARR** (four quarters ahead), and
- Growth diagnostics (QoQ and YoY rates) and uncertainty bounds.

Internally, the core model predicts **YoY ARR growth rates for each of the next four quarters** and converts those into absolute ARR by applying predicted YoY growth to last year's corresponding quarter.

0.2.2 1.2 Real-world constraint: minimal user input

The training dataset contains many features per company-quarter, but a real VC workflow cannot assume that an investor (or founder) will provide that full feature set. The central engineering requirement is therefore:

> Enable high-quality predictions from minimal inputs without producing feature vectors that are out-of-distribution relative to training data.

0.2.3 1.3 Success criteria (VC + ML)

- **Accuracy**: strong predictive fit on held-out data (R^2/MAE), with horizon-wise reporting.
- **Robustness under missingness**: stable results with Tier 1 only; improved fidelity with Tier 2.
- **Behavior on edge cases**: declining, flat, and volatile companies should not receive unrealistic "high-growth" forecasts.
- **Explainability**: routing decisions and rule-based projections must be auditable for investment decision support.
- **Operational readiness**: API deployment, validation, error handling, and latency appropriate for interactive use.

0.3 2. Data Overview

0.3.1 2.1 Dataset summary (production dataset)

The production model uses `202402_Copy.csv`:

- **Observations**: 5,085 company-quarters
- **Companies**: 354
- **Raw columns**: 122

This dataset is **not iid**: it contains repeated measures per company and reflects venture-backed dynamics (high growth, volatility, down periods, and reporting inconsistency).

0.3.2 2.2 Bias and representativeness

The dataset is geographically concentrated (US and a small set of other ecosystems dominate). A representative snapshot is shown in the country distribution figure.

Figure 4. Top countries in training data.

Sectors are also concentrated in software/tech verticals. The production API further constrains sectors to a consolidated list for validation and robustness.

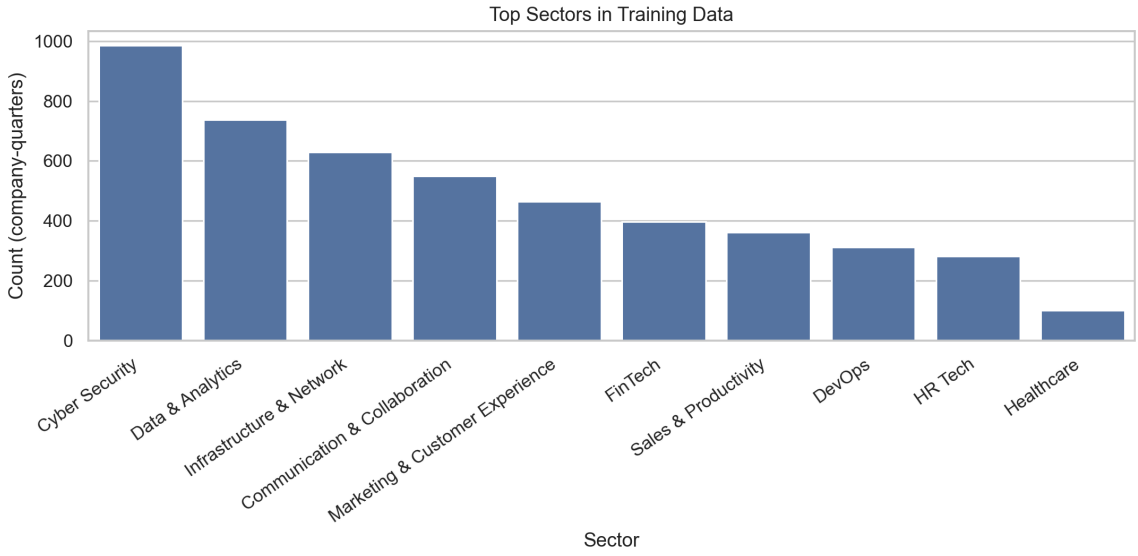

Figure 5. Top sectors in training data.

0.3.3 2.3 Missingness and why it matters

Many features are sparse (often missing in 80–90%+ of rows). This strongly influences both model training and system design.

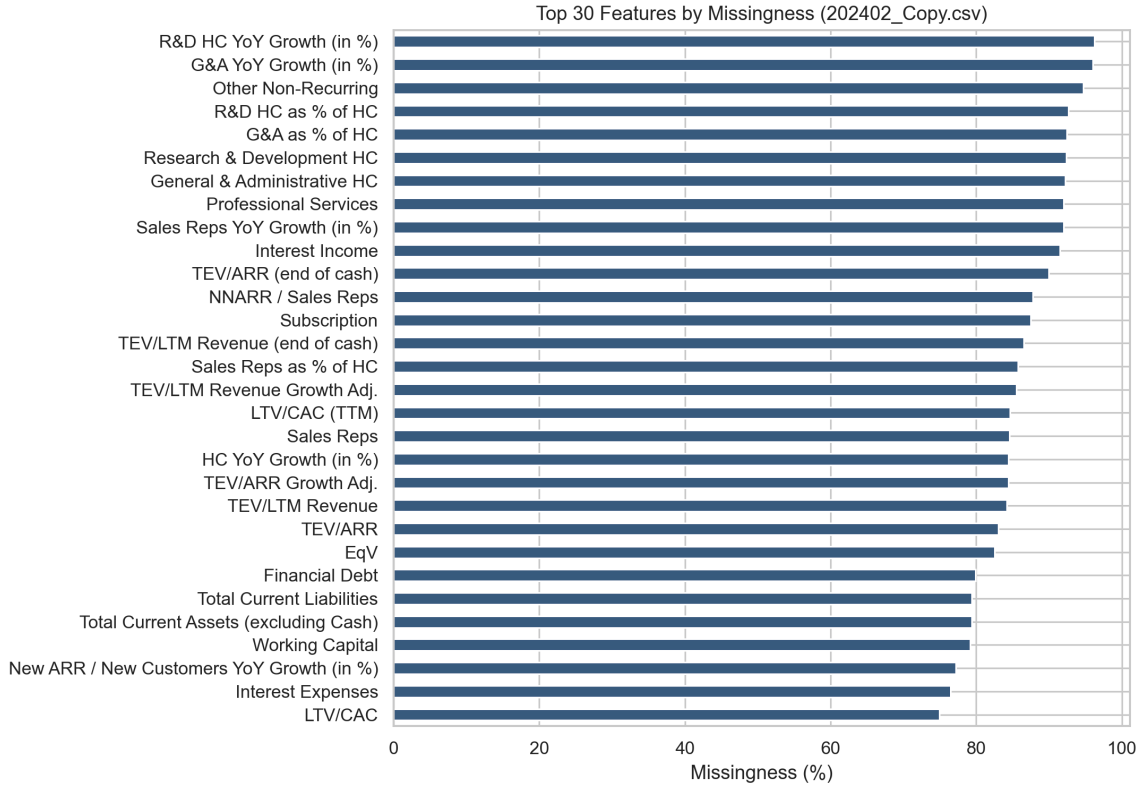

Figure 1. Top-30 feature missingness in training data.

Key implication: naïve “fill missing with 0” at inference creates unrealistic inputs and causes distribution shift. This observation directly motivated **Intelligent Feature Completion**.

0.3.4 2.4 Heavy tails and volatility

ARR and growth metrics are heavy-tailed, spanning orders of magnitude. This motivates:

- logarithmic similarity for peer matching,
- robust feature imputation (weighted medians), and
- a hybrid path for non-standard trajectories.

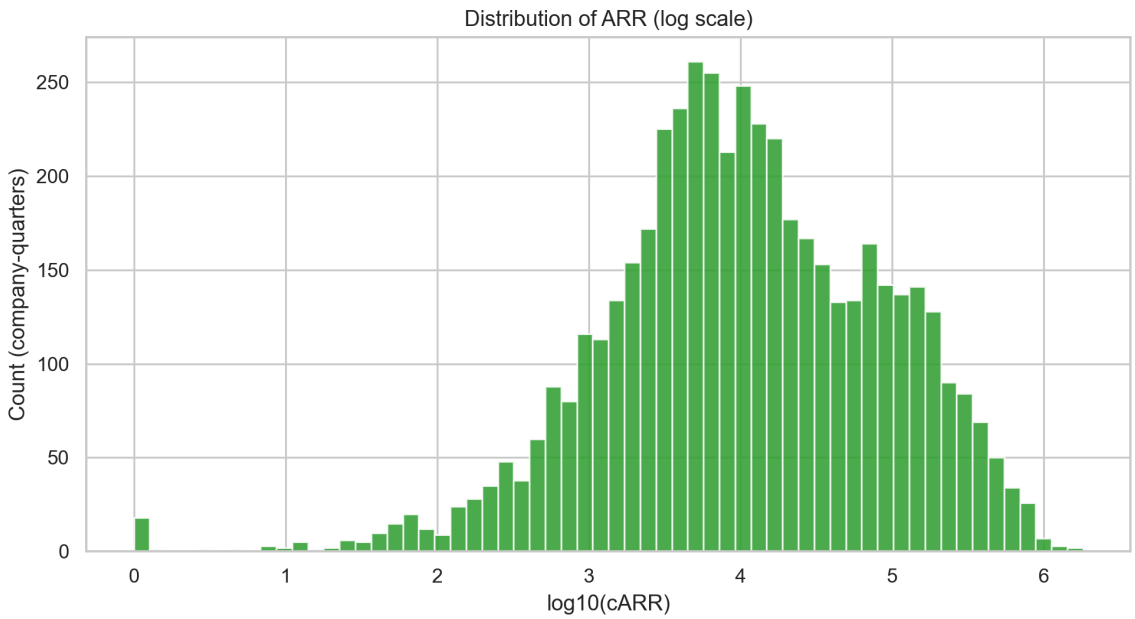

Figure 2. ARR distribution (log scale).

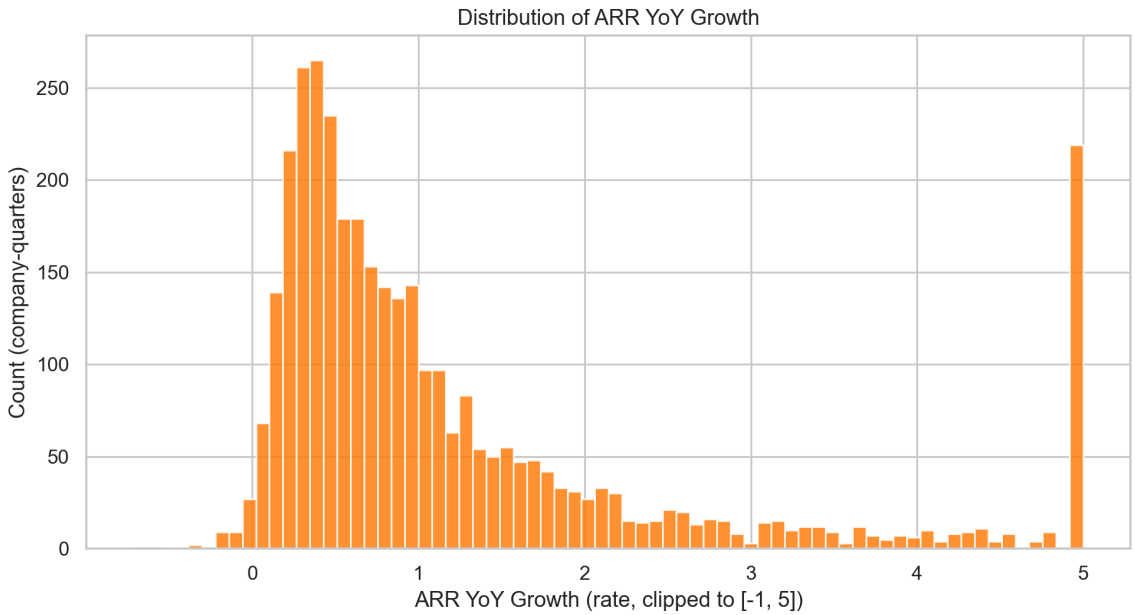

Figure 3. ARR YoY growth distribution (clipped for readability).

0.4 3. System Architecture (End-to-End)

0.4.1 3.1 Production dataflow

The production API integrates a small number of “active” modules:

1. Request validation (FastAPI + Pydantic)
2. Tier-based input parsing
3. Trend detection and routing
4. Either:
    - A. ML route: intelligent feature completion → LightGBM prediction, or
    - B. Edge-case route: rule-based health assessment projection
5. Post-processing: QoQ, YoY, scenario bounds (±10%), response formatting

0.4.2 3.2 API surface

Core endpoints:

- `POST /tier_based_forecast`: main forecast endpoint
- `POST /predict_csv`: batch forecasts from CSV upload
- `POST /chat`: conversational explanation + analysis tools
- `GET /makro-analysis`: macro indicators (VIX/MOVE/BVP/GPRH)
- `GET /health`, `GET /model_info`: system diagnostics

0.5 4. Tier-Based Input Contract

0.5.1 4.1 Tier 1 (required)

Tier 1 is designed for the minimum information that is typically obtainable during diligence:

- Q1–Q4 ARR (most recent four quarters)
- headcount
- sector (constrained to a validated set)

0.5.2 4.2 Tier 2 (optional)

Tier 2 allows a user to increase forecast fidelity where available:

- gross margin
- sales & marketing spend
- cash burn
- customers
- churn rate
- expansion rate

This tiering prevents the system from failing when a metric is missing while still rewarding higher-quality inputs.

0.6 5. Intelligent Feature Completion (Core Contribution)

0.6.1 5.1 Why feature completion is necessary

The LightGBM model expects a full engineered feature vector. If a user provides only Tier 1, most features would be missing at inference. Simple imputation (zeros/global means) produces vectors that are not representative of the training distribution, harming performance and trust.

0.6.2 5.2 Similarity matching

The completion system finds a cohort of similar companies using:

- **logarithmic similarity on ARR** (robust to scale),
- similarity on growth proxies, and
- similarity on headcount.

It then selects the **top 50** nearest companies as a peer set.

0.6.3 5.3 Robust inference via weighted median

For each missing feature, values are inferred from the peer set using a **weighted median** (weights = similarity score). This improves robustness under heavy-tailed feature distributions typical in startup metrics.

0.6.4 5.4 Default priors by ARR scale

If a feature is absent even in the peer set (or fails type conversions), the system falls back to size-dependent defaults (small/growth/large ARR regimes). This avoids pathologies such as negative headcount or implausible margins.

0.7 6. Hybrid Forecasting: ML Route + Rule-Based Health Assessment

0.7.1 6.1 Why a hybrid system exists

A pure ML system trained predominantly on growth regimes can produce unrealistic outputs for:

- consistent decline,
- flat/stagnant revenue,
- highly volatile patterns, and
- trend reversals.

This is a classic distribution shift problem: the ML model has limited exposure to these regimes and is optimized for typical growth dynamics.

0.7.2 6.2 Trend detection and routing (6-factor analysis)

A dedicated routing layer analyzes the last four quarters using:

- overall Q1→Q4 growth,
- QoQ sequence,
- recent momentum (Q3→Q4),
- consistency,
- volatility,
- acceleration/deceleration.

This generates a trend label (e.g., CONSISTENT_GROWTH, CONSISTENT_DECLINE, VOLATILE_IRREGULAR) and determines whether to route to the edge-case method.

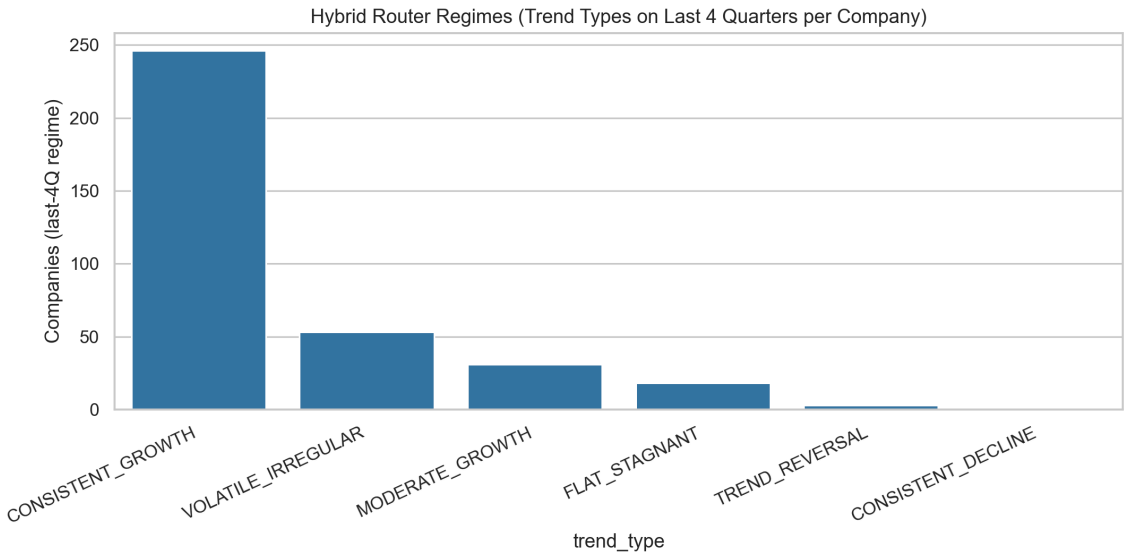

Figure 9. Trend-type distribution (proxy: last-4Q per company).

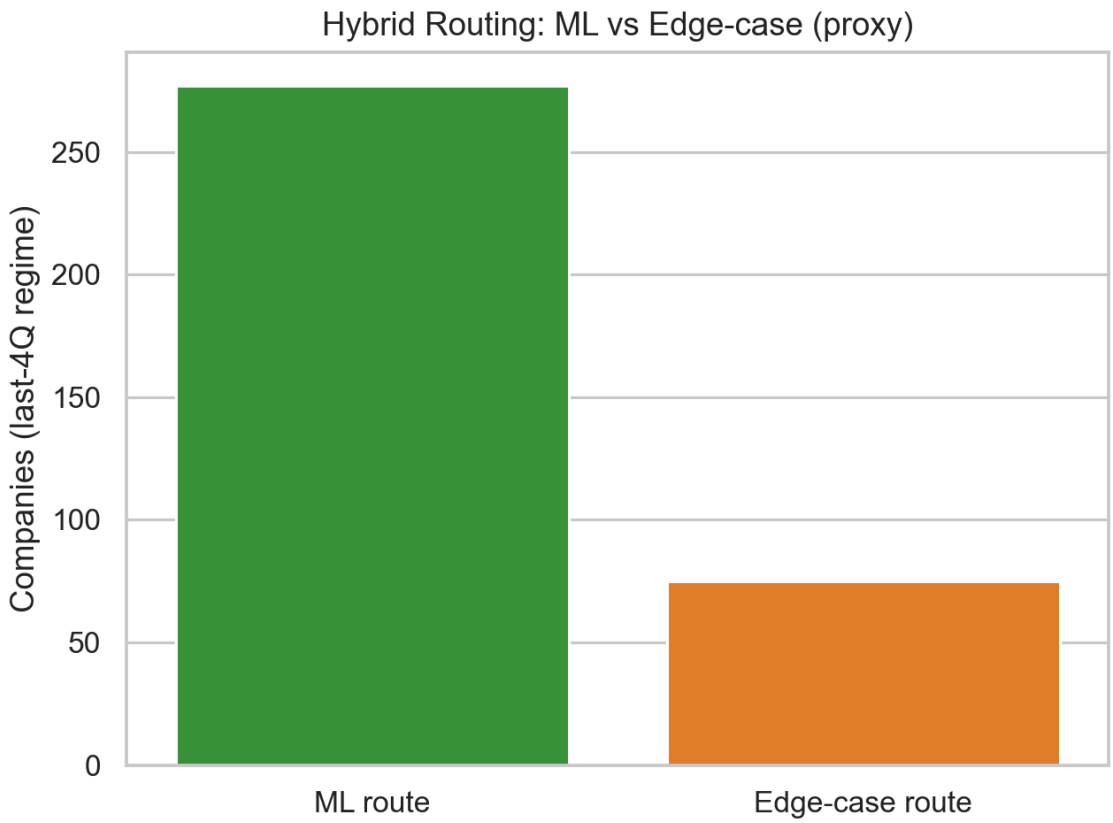

Figure 10. Hybrid routing breakdown (proxy).

0.7.3 6.3 Rule-based health assessment (edge-case method)

When the router detects an edge case (or Tier 2 indicates poor fundamentals such as short runway), the system uses a deterministic, benchmark-driven method that:

- computes health metrics (ARR growth, NRR, CAC payback, Rule of 40, runway),
- assigns a health tier (HIGH / MODERATE / LOW), and
- projects ARR forward using conservative growth/decline rules.

This yields an explainable forecast suitable for investment committees and post-investment monitoring.

0.8 7. Model: LightGBM Multi-Output Forecasting

0.8.1 7.1 Model choice

LightGBM is chosen due to:

- strong performance on tabular data,
- ability to capture nonlinear interactions,
- efficient inference for API serving,
- robustness under engineered features.

0.8.2 7.2 Forecast horizons and evaluation

Model performance by horizon:

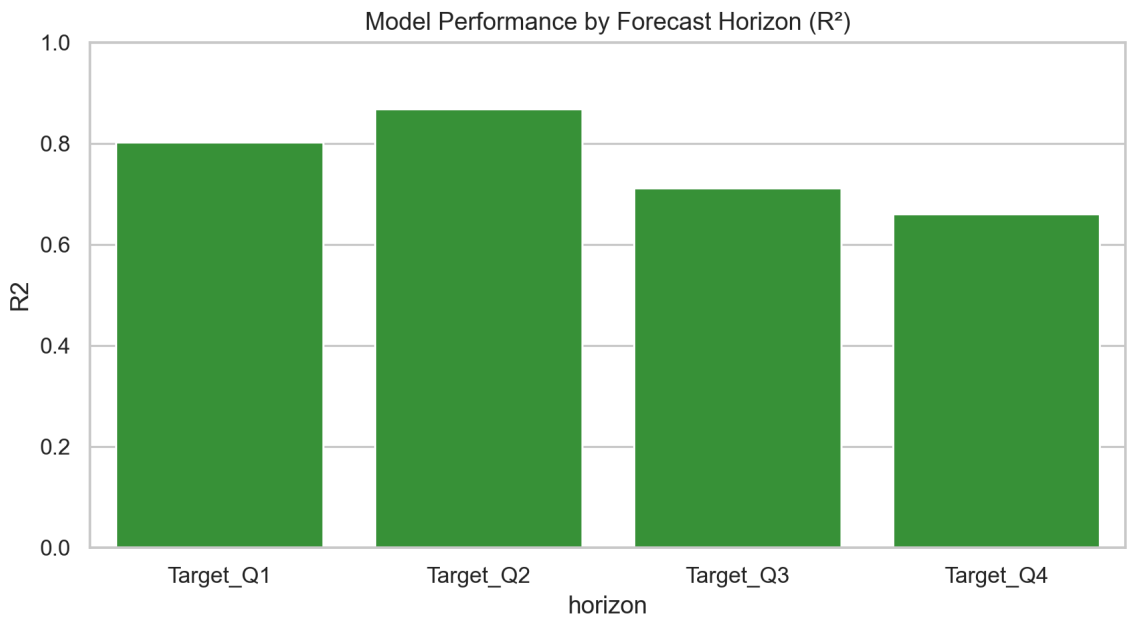

Figure 6. R² by forecast horizon.

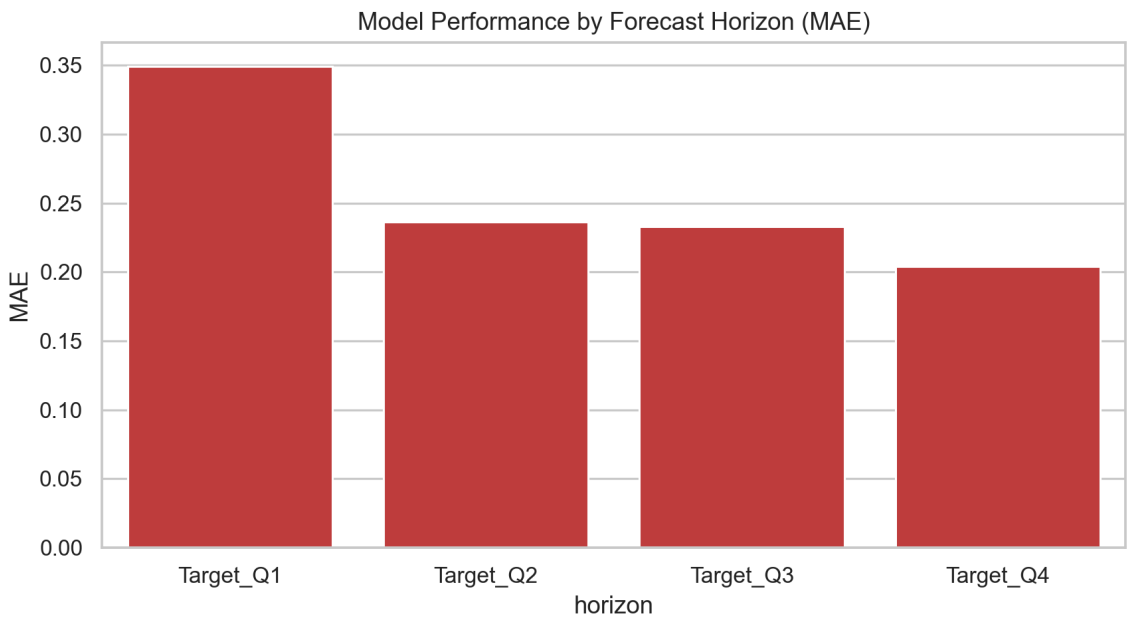

Figure 7. MAE by forecast horizon.

0.8.3 7.3 Feature importance

The top importance features provide partial interpretability and sanity checks (e.g., ARR growth and efficiency metrics should matter in SaaS).

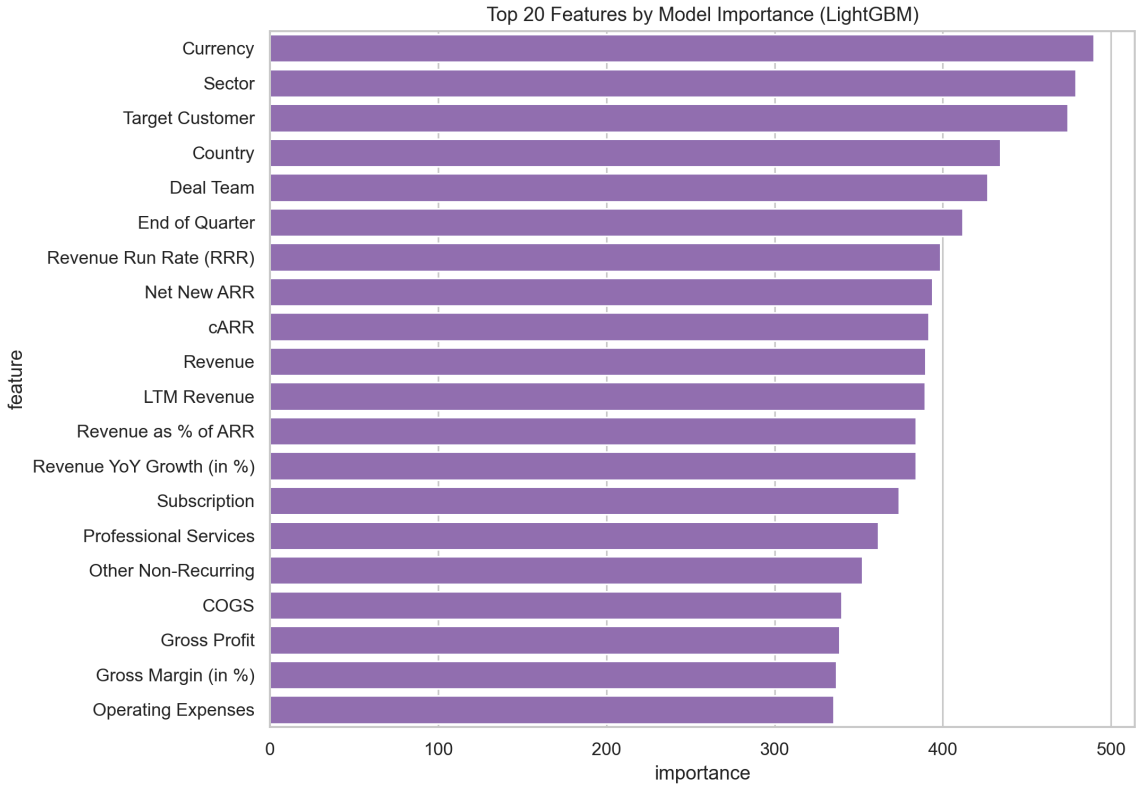

Figure 8. Top 20 features by model importance.

0.9 8. Uncertainty Quantification (Decision-Support Range)

All forecasts return three scenarios:

- pessimistic: (-10%)
- realistic: model projection
- optimistic: (+10%)

This is implemented as a pragmatic **scenario band**, not a calibrated statistical prediction interval. The intent is to provide:

- a risk-aware range for VC decision-making, and
- a consistent communication format for founders and investors.

Future iterations should calibrate this band based on historical residual distributions (e.g., empirical 10th/90th percentile errors by ARR regime and horizon).

0.10 9. Macroeconomic Context Module

The system provides macro indicators via `GET /makro-analysis`:

- **VIX**: equity volatility / risk sentiment
- **MOVE**: rates volatility (funding conditions)
- **BVP Cloud Index**: public cloud/software valuation proxy
- **GPRH**: geopolitical risk

These indicators are used as **context**, not as causal features in the core ARR model. The goal is to present the forecast alongside market regime information, improving interpretation and decision quality.

0.11 10. Limitations and Threats to Validity

- **Sampling bias**: dataset is concentrated in specific geographies and software sectors; generalization is strongest for similar companies.
- **Missingness**: many metrics are not consistently reported; imputation introduces uncertainty and potential bias.
- **Heuristic uncertainty**: ±10% band is not calibrated; it should be validated per segment/horizon.
- **Regime shift**: model performance may degrade under structural changes (rates, GTM shifts, competitive dynamics).
- **Similarity assumptions**: “nearest companies” may not reflect true similarity for unique businesses; the system can inherit peer group bias.

0.12 11. Conclusion

This project delivers a production-ready ARR forecasting system for venture capital workflows. The core contributions are:

- a tiered input interface that matches real diligence constraints,
- a similarity-driven feature completion approach to prevent out-of-distribution inference,
- a hybrid routing architecture that prevents unrealistic forecasts for edge cases via explainable, benchmark-driven projections,
- a deployed FastAPI surface including batch prediction, chat-based explanations, and macro context.

The system demonstrates strong predictive performance ((R^2)) while prioritizing robustness and interpretability—two requirements that determine whether ML forecasting tools are usable in real investment practice.

0.12.1 Appendix A. How to reproduce figures

From the repo root:

```
python3 report/generate_figures.py
```

Figures are written to `report/figures/`.